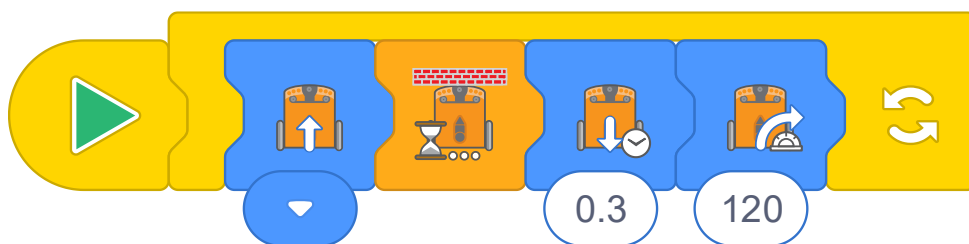




EdBlocks block guide

Guide to graphical block programming with the Edison robot



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www.meetedison.com

Contents

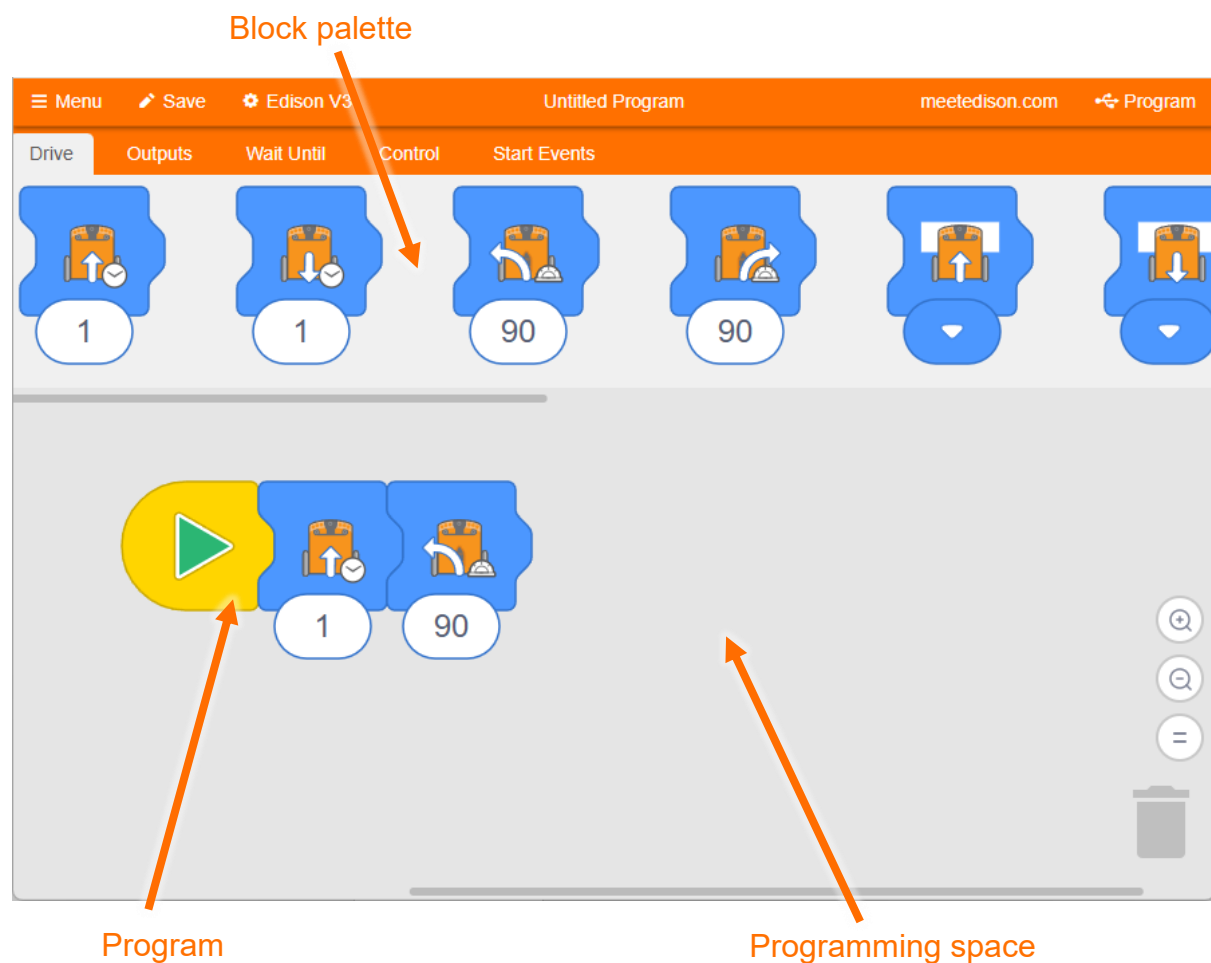
About EdBlocks.....	3
Drive for time.....	4
Turn for degrees	5
Set drive	6
Drive while on white surface	7
Drive while on black surface	8
Drive while light on left side	9
Drive while light on right side	10
Follow a line	11
Stop drive.....	12
Set drive speed	13
LED lights.....	14
Musical notes	15
Next note sharp.....	18
Set music tempo	19
Send message	20
Wait until time	21
Wait until clap.....	22
Wait until button press	23
Wait until surface (black or white).....	24
Wait until light.....	25
Wait until obstacle	26
Wait until message	27
Loop forever.....	28
Loop number of times	29
Loop until button pushed	30
Loop until obstacle	31
End program	32
Start when obstacle detected	33
Start when clap detected	34
Start when button pushed	35
Start when on surface	36
Start when message received	37
Start when remote command received.....	38
Remote commands.....	39



About EdBlocks

EdBlocks is the entry level block programming language for the Edison V3 robot. You can find the app here: edblocksapp.com

Using EdBlocks, programs are constructed by dragging blocks from the block palette and placing them into the programming space. Programs are made up of blocks connected to yellow Start Events blocks. Blocks not connected to yellow Start Events blocks are not part of the program and are not downloaded into Edison.



About this block guide

Use this guide to understand what each block does and as a reference for the example programs.

You can find more information about EdBlocks, including lessons here: <https://meetedison.com/robot-programming-software/edblocks/>



Drive for time

The 'drive for time' blocks drive the Edison robot for the set number of seconds.

There are three speed settings that are controlled with the 'set drive speed' block.

Hint: If the Edison is not driving straight, perform a calibration here:

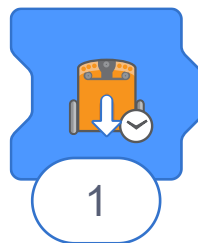
<https://meetedison.com/turn-calibrate-the-edison-v3-robot/>

There are two 'drive for time' blocks:

Drive forward

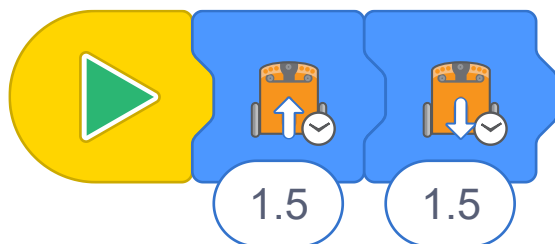


Drive backward



Example program

The Edison robot drives forwards for 1.5 seconds and then backwards for 1.5 seconds.



Turn for degrees

The 'turn for degrees' blocks drive the Edison robot to turn for a number of degrees.

When turning right, the Edison drives the left wheel and when turning left the Edison drives the right wheel.

Hint: If the Edison is not turning accurately, perform a calibration here:

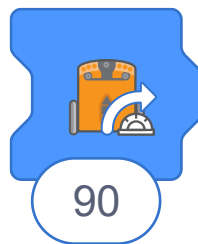
<https://meetedison.com/turn-calibrate-the-edison-v3-robot/>

There are two 'turn for degrees' blocks:

Turn left

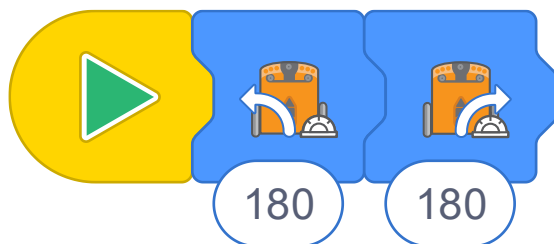


Turn right



Example program

The Edison robot turns left 180 degrees and then right 180 degrees.



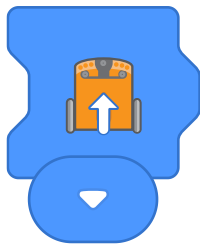
Set drive

The 'set drive' blocks set the Edison robot to drive. The robot will continue to drive until another drive block either stops the robot or changes its direction.

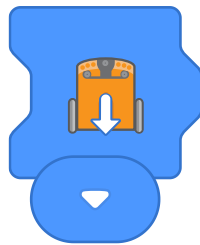
Additional options for the 'set drive' blocks can be found by clicking the dropdown arrow.

There are four 'set drive' blocks:

Drive forward



Drive backward



Turn left

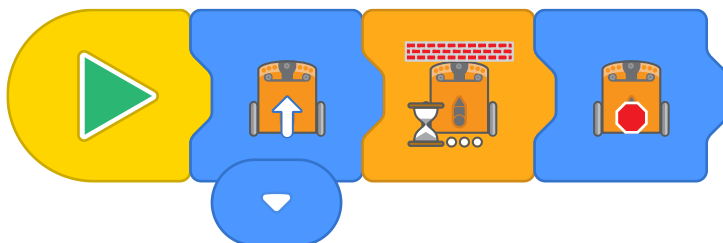


Turn right



Example program:

The Edison robot is set to drive forward, then the program pauses until an obstacle is detected, when this happens the program progresses to the 'stop drive' block and the Edison robot stops driving. This is a stop at obstacle program.



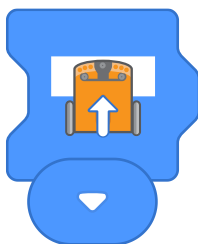
Drive while on white surface

The 'drive while on white surface' blocks set the Edison robot to drive while on a white surface. The line tracking detector is used to sense whether the surface the robot is driving on is black or white.

The program will pause at this block and the robot will drive until the surface is no longer white. When the driving surface changes to black, the robot stops driving and the program progresses to the next block.

There are four 'drive while on white' blocks:

Drive forward
while on white



Drive backward
while on white



Turn left while
on white

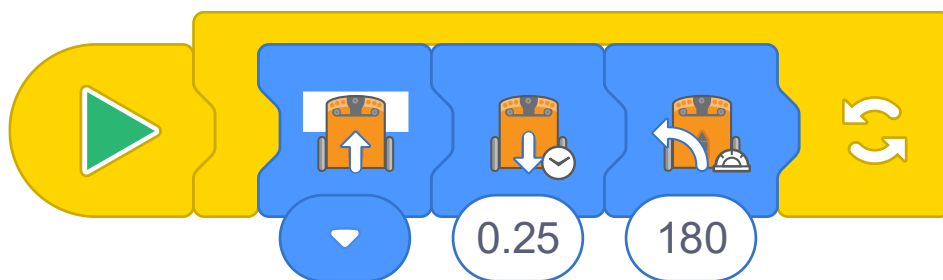


Turn right while
on white



Example program

The Edison robot is set to drive forward while on a white surface, when the surface changes to black the program progresses to drive backwards for 0.25 seconds, then turn left 180 degrees. These blocks are inside a 'loop forever' block, so the program repeats forever. This is a simple bounce in borders program. To test the program, place the Edison robot inside a ring with a thick black border.



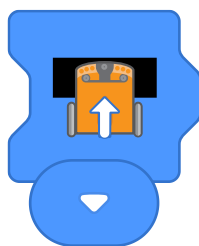
Drive while on black surface

The 'drive while on black surface' blocks set the Edison robot to drive while on a black surface. The line tracking detector is used to sense whether the surface the robot is driving on is black or white.

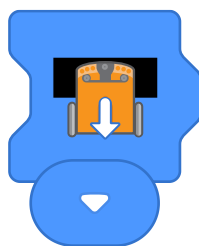
The program will pause at this block and the robot will drive until the surface is no longer black. When the driving surface changes to white, the robot stops driving and the program progresses to the next block.

There are four 'drive while on black' blocks:

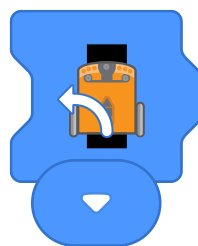
Drive forward while on black



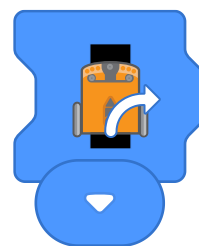
Drive backward while on black



Turn left while on black

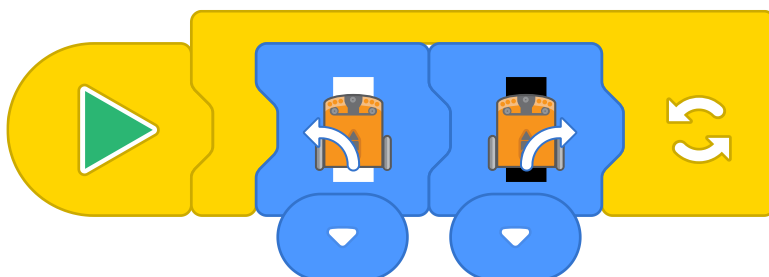


Turn right while on black



Example program

The Edison robot is set to drive left while on a white surface and then drive right on a black surface. The blocks are inside a 'loop forever' block, so the program repeats forever. This is a simple line following program. To test the program, place the robot next to a thick black line.



Drive while light on left side

The 'drive while light on left side' blocks set the Edison robot to drive while the light on the left light sensor is brightest (brighter than the right light sensor). The light sensors on each side of the Edison robot are used to sense the amount of light.

The program will pause at this block and the robot will drive until the light is no longer brightest on the left light sensor. When the light level on the left light sensor is no longer the brightest, the robot stops driving and the program progresses to the next block.

There are four 'drive while light on left side' blocks:

Drive forward
while light on
left side



Drive backward
while light on
left side



Turn left while
light on left side

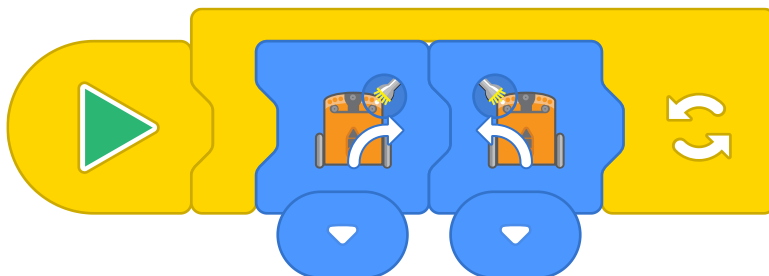


Turn right while
light on left side



Example program

The Edison robot is set to drive right when the light is brightest on the right, then when the light is brightest on the left, it is set to drive left. Both blocks are inside a 'loop forever' block, so the program repeats forever. This program is a light following program. To test it use a flash light and point it at the Edison robot.



Drive while light on right side

The 'drive while light on right side' blocks set the Edison robot to drive while the light on the right light sensor is brightest (brighter than the left light sensor). The light sensors on each side of the Edison robot are used to sense the amount of light.

The program will pause at this block and the robot will drive until the light is no longer brightest on the right light sensor. When the light level on the right light sensor is no longer the brightest, the robot stops driving and the program progresses to the next block.

There are four 'drive while light on right side' blocks:

Drive forward
while light on
right side



Drive backward
while light on
right side



Turn left while
light on right side

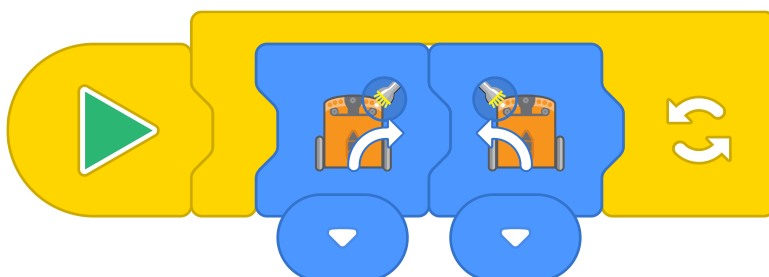


Turn right while
light on right side



Example program

The Edison robot is set to drive right when the light is brightest on the right, then when the light is brightest on the left, it is set to drive left. Both blocks are inside a 'loop forever' block, so the program repeats forever. This program is a light following program. To test it use a flash light and point it at the Edison robot.



Follow a line

The 'follow a line' blocks set the Edison robot to follow a black line.

Hint: Always start with the Edison robot next to the black line, not on the black line (i.e. on white).

There are three 'follow a line' blocks:

Follow a line
for time



Follow a line
until an obstacle

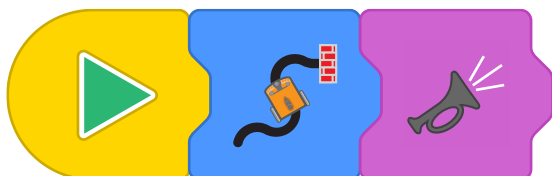


Follow a line
forever



Example program

The Edison robot follows a black line until an obstacle is detected, stops and then plays a beep sound.



Stop drive

The 'stop drive' block stops the Edison robot from driving.

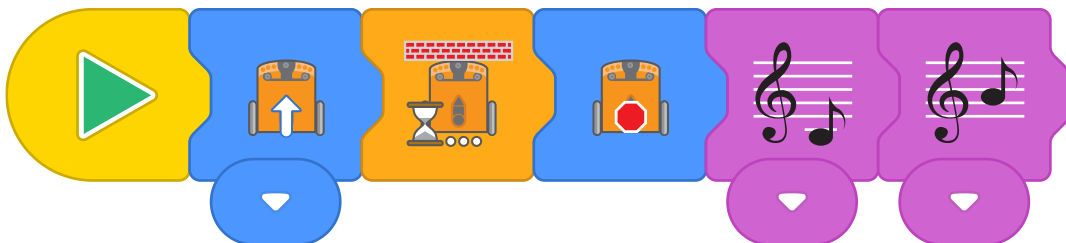
When the program runs this block both wheels stop driving.

This is the 'stop drive' block:



Example program

The Edison robot is set to drive forward, the program then pauses until an obstacle is detected, when this happens the stop drive block stops the robot from driving and two notes are played before the program ends.



Set drive speed

The 'set drive speed' block controls the speed that the Edison robot drives at. Use the dropdown arrow to see all of the block options.

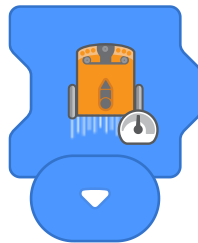
When the program runs this block, all drive blocks after this block will have their speed adjusted to the speed setting.

There are three speed settings:

Set speed
slow



Set speed
medium

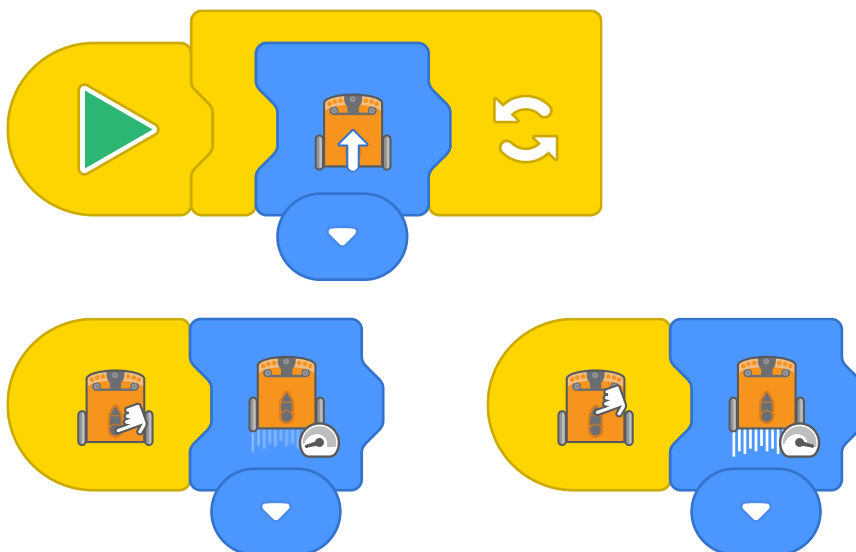


Set speed
fast



Example program

The Edison robot is set to drive forward inside a 'loop forever' block. When the round button is pressed the drive speed is set to slow. When the triangle button is pressed the drive speed is set to fast.

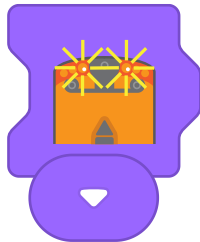


LED lights

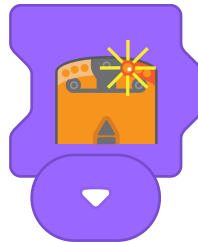
The 'LED lights' blocks control turning on and turning off the LED lights. Use the dropdown arrow to see all of the block options.

There are six 'LED lights' blocks:

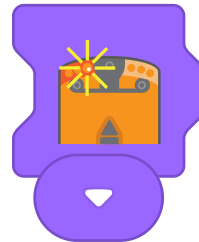
Both LEDs on



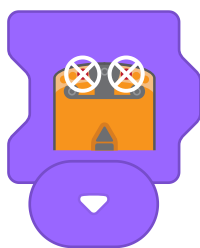
Right LED on



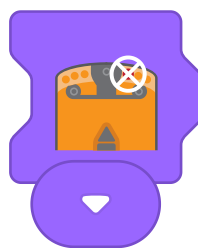
Left LED on



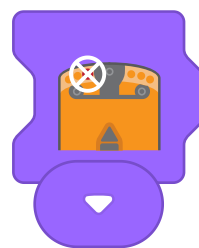
Both LEDs off



Right LED off

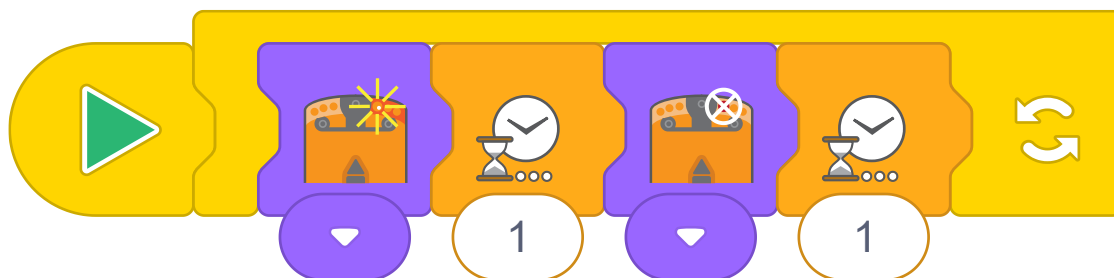


Left LED off



Example program

The right LED is turned on for one second and then turned off for one second. These blocks are inside a 'loop forever' block, so the LED will flash forever.



Musical notes

The 'musical note' blocks play musical notes via the Edison robot's speaker. Use the dropdown arrows to see all of the note options.

There are forty musical note blocks:

whole C



half C



quarter C



eighth C



whole B



half B



quarter B



eighth B



whole A



half A



quarter A



eighth A



whole G



half G



quarter G



eighth G



whole F



half F



quarter F



eighth F



whole E



half E



quarter E



eighth E



whole D



half D



quarter D



eighth D



whole middle C



half middle C



quarter middle C



eighth middle C



whole low B



half low B



quarter low B



eighth low B



whole rest



half rest



quarter rest



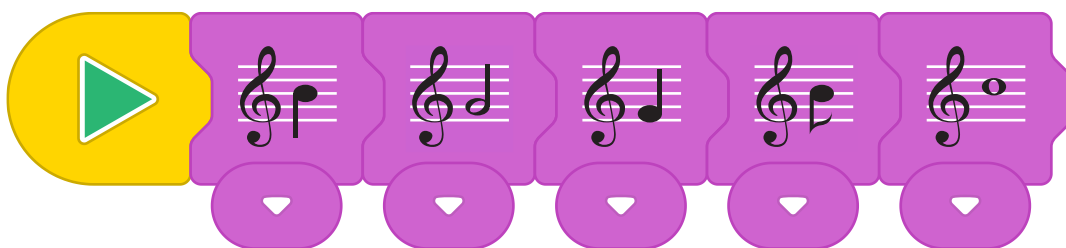
eighth rest



Example program

The Edison plays the following notes:

quarter B, half G, quarter F, eighth B, whole C.



Next note sharp

The 'next note sharp' block sets any musical note that follows this block to be played a sharp note.

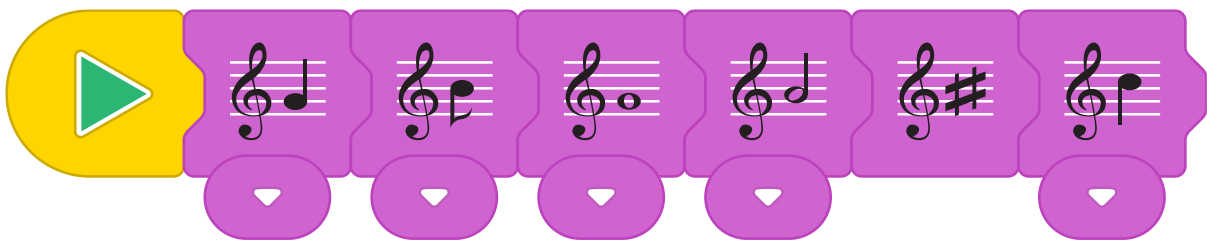
This is the 'next note sharp' block:



Example program

The Edison plays the following notes:

quarter G, eighth B, whole G, half A, quarter C#



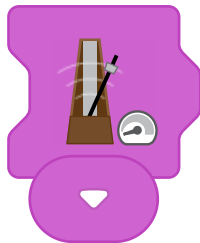
Set music tempo

The 'set music tempo' block sets the music tempo that the musical blocks in a program are played at. If the tempo block is not used, then the tempo is normal. Use the dropdown arrow see all of the block options.

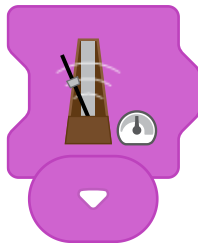
Hint: This block should always be used at the beginning of a program.

There are three 'set music tempo' blocks:

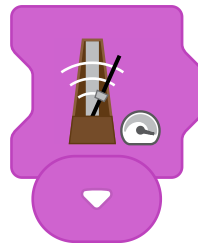
Set music
tempo to slow



Set music
tempo to normal



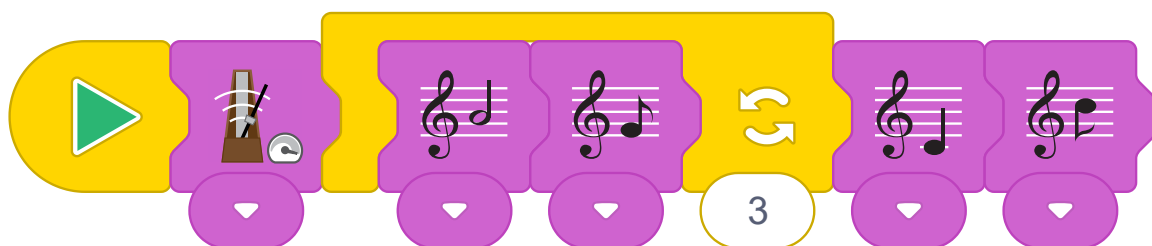
Set music
tempo to fast



Example program

The Edison plays the following notes at fast tempo:

half A, eighth F, half A, eighth F, half A, eighth F, quarter middle C, eighth C



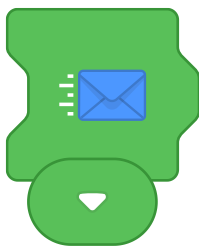
Send message

The 'send message' blocks send a coloured message via infrared to another Edison robot. Use the dropdown arrow to select the different messages. Use the dropdown arrow see all of the block options.

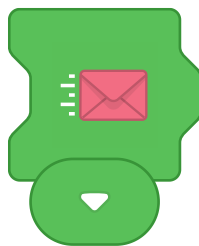
The receiving Edison robot needs to be programmed using the 'wait until message received' block or a 'message received start event' block.

There are six 'send message' blocks:

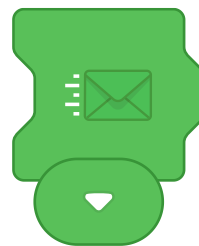
Send blue message



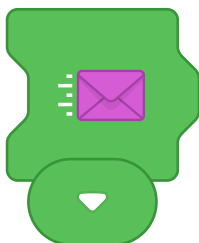
Send pink message



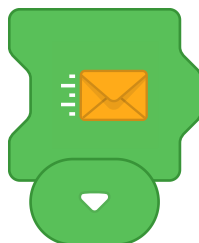
Send green message



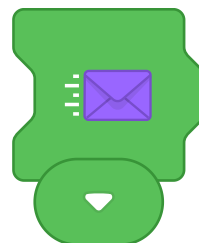
Send violet message



Send orange message



Send indigo message



Example program

The Edison waits until the round button is pressed, when this happens the program progresses to the next block and sends a blue message. These blocks are inside a 'loop forever' block, so the program repeats forever. This program sends a blue message anytime the round button is pressed.



Wait until time

The 'wait until time' block pauses the program until the amount of time in seconds has passed.

The minimum amount of time that the block will wait is 0.01 seconds.

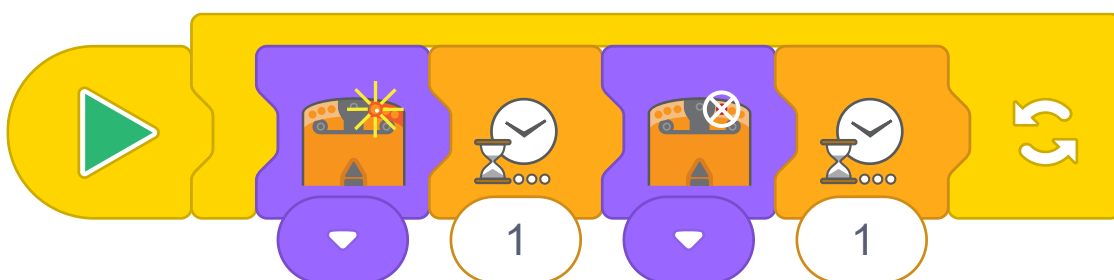
The maximum amount of time that the block will wait is 9,999 seconds (almost 3 hours).

This is the 'wait until time' block:



Example program

The right LED is turned on for one second and then turned off for one second. The blocks are inside a 'loop forever' block, so the LED will flash forever.



Wait until clap

The 'wait until clap' block pauses the program until the set number of claps have been detected.

There are two options for the number of claps detected; either one clap or two claps.

Use the dropdown arrow to select the 'wait until clap' block that waits until either one or two claps.

There are two 'wait until clap' blocks:

Wait until
one clap

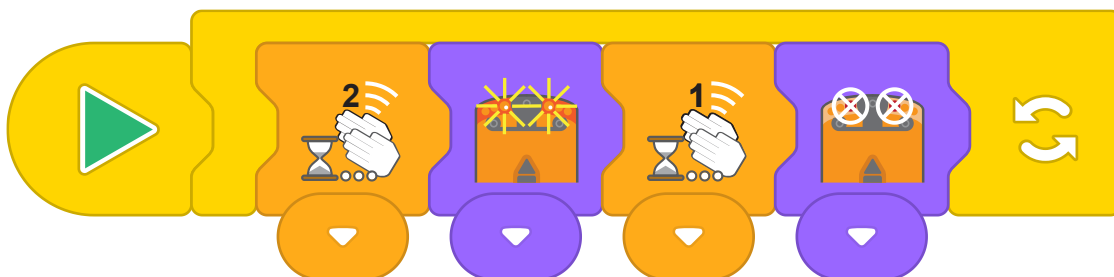


Wait until
two claps



Example program

The Edison waits until two claps are detected, when this happens the program progresses to the next block and turns on both LED lights. The Edison now waits for just one clap. When this happens both LED lights are turned off. These blocks are inside a 'loop forever' block, so the program repeats forever. This program turns on Edison's LEDs when two claps are detected and turns them off when one clap is detected.



Wait until button pushed

The 'wait until button pushed' block pauses the program until the selected button is pushed.

There are two options for which button is pushed; either the round button or the triangle button.

Use the dropdown arrow to select the 'wait until button pushed' block for either the round button or triangle button.

There are two 'wait until button pushed' blocks:

Wait until round
button pushed

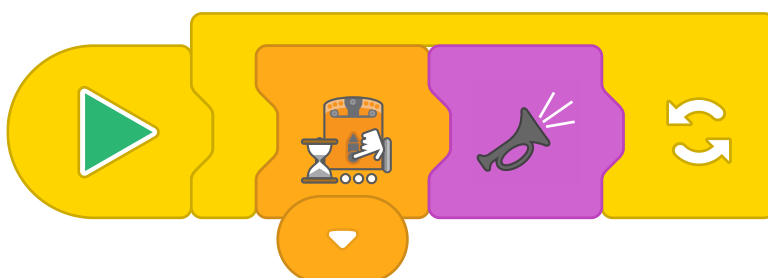


Wait until triangle
button pushed



Example program

The Edison waits for the round button to be pushed. When this happens the program progresses to the next block. This block is the beep block and Edison beeps. These blocks are inside a 'loop forever' block, so the program repeats forever. This program makes Edison beep whenever the round button is pressed.



Wait until surface (black or white)

The 'wait until surface' block pauses the program until the line tracker sensor detects either a black or a white driving surface.

Use the dropdown arrow to select the 'wait until surface' block for either the black or white surface.

There are two 'wait until surface' blocks:

Wait until
black surface

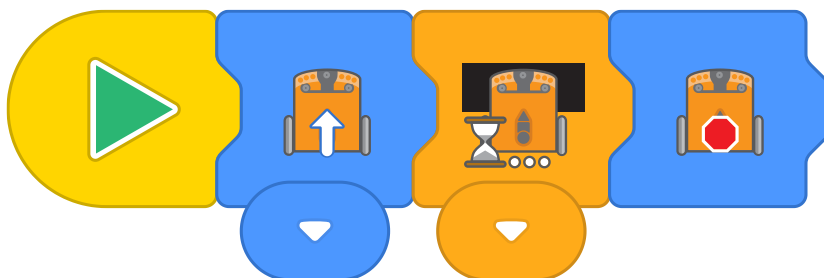


Wait until
white surface



Example program

The Edison robot is set to drive forward, then waits until a black surface is detected. When this happens the program progresses to stop driving and the program ends.



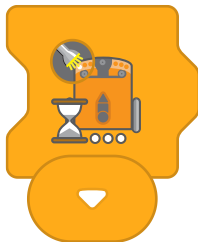
Wait until light

The 'wait until light' block pauses the program until the light on the left or the right side is brightest.

Use the dropdown arrow to select the 'wait until light' block for either the light on left or light on right.

There are two 'wait until light' blocks:

Wait until
light on left

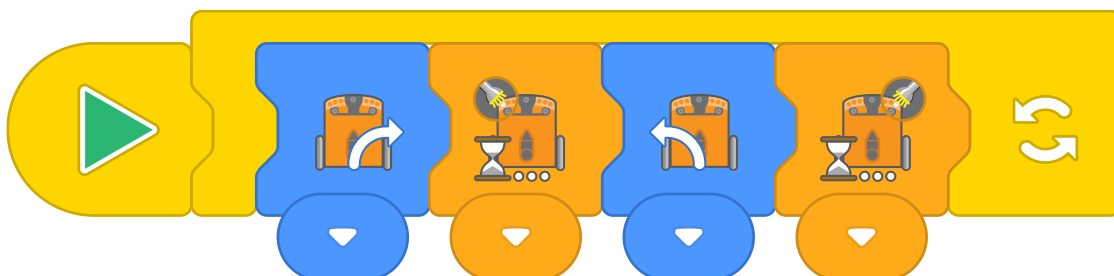


Wait until
light on right



Example program

The Edison robot drives right until the light is brightest on the left side. When this happens the program progresses to the next block. Now, Edison drives left until the light is brightest on the right side. These blocks are inside a 'loop forever' block, so the program repeats forever. This program makes Edison drive towards bright light.



Wait until obstacle

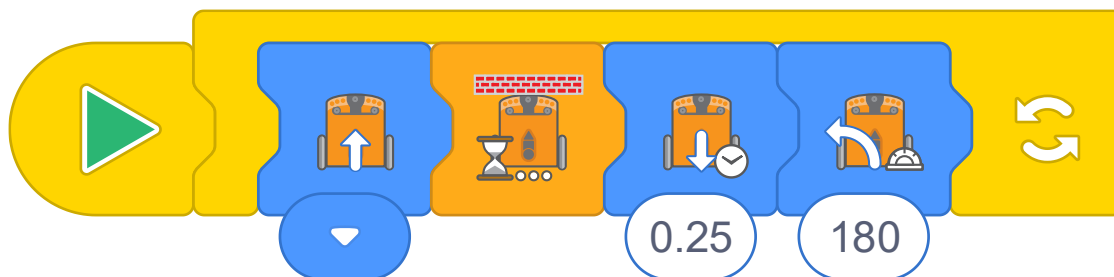
The 'wait until obstacle' block pauses the program until an obstacle is detected.

This is the 'wait until obstacle' block:



Example program

The Edison robot drives forward until an obstacle is detected. When this happens the program progresses to the next block. Now the Edison robot drives backwards for 0.25 seconds, then turns 180 degrees. These blocks are inside a 'loop forever' block, so the program repeats forever. This program makes Edison back up and turn away from obstacles in its path.



Wait until message received

The 'wait until message received' block pauses the program until the selected colour message is received. Use the dropdown arrow see all of the block options.

There are six 'wait until message received' blocks:

Wait until blue message received



Wait until pink message received



Wait until green message received



Wait until violet message received



Wait until orange message received

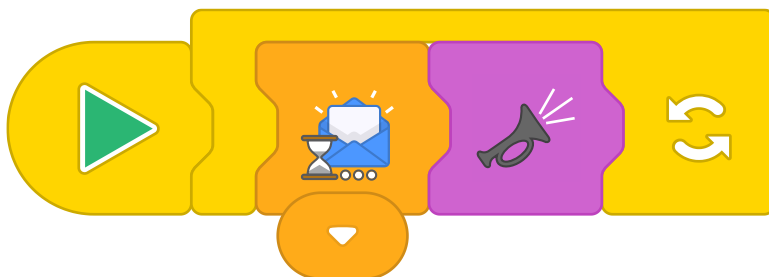


Wait until indigo message received



Example program

The Edison waits until a blue message is received from another Edison robot. When this happens the program progresses to the next block and plays a beep. These blocks are inside a 'loop forever' block, so the program repeats forever. This program makes Edison beep whenever a blue message is received.



Loop forever

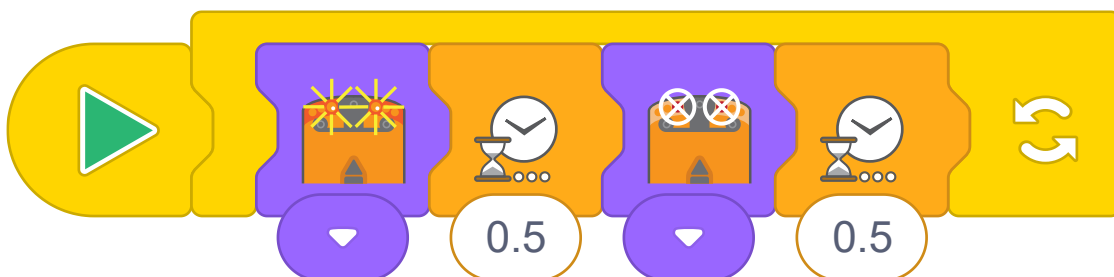
The 'loop forever' block repeats all the blocks placed inside of it, forever.

This is the 'loop forever' block:



Example program

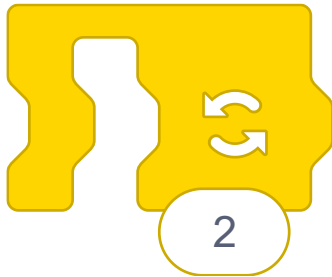
The Edison turns both LED lights on, then waits until 0.5 seconds have passed, then turns both LEDs off and waits until 0.5 seconds have passed. These blocks are placed inside a 'loop forever' block so they will repeat forever.



Loop number of times

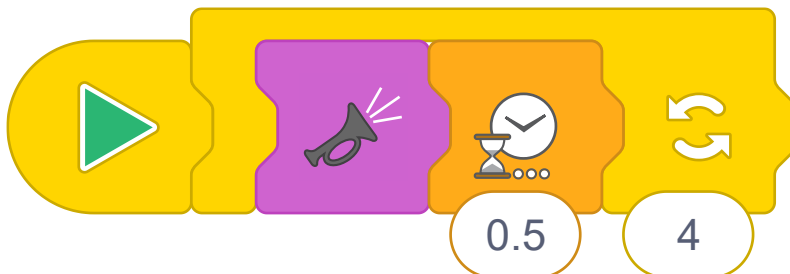
The 'loop number of times' block repeats all of the blocks placed inside of it for the specified number of times.

This is the 'loop number of times' block:



Example program

The Edison robot plays a beep, then waits for 0.5 seconds. These blocks are inside a 'loop number of times' block set to 4 times, so Edison will play a beep sound four times.

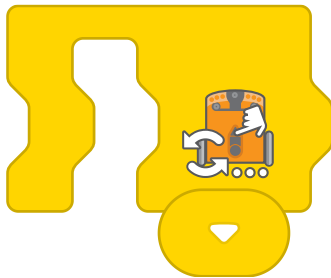


Loop until button pushed

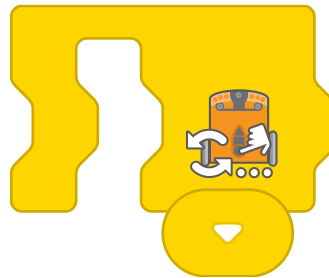
The 'loop until button pushed' blocks repeat the blocks placed inside of it until the specified button is pushed. Use the dropdown arrow see all of the block options.

There are two 'loop until button pushed' blocks:

Loop until triangle button pushed

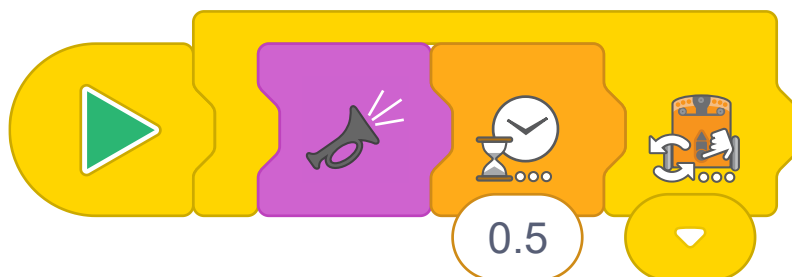


Loop until round button pushed



Example program

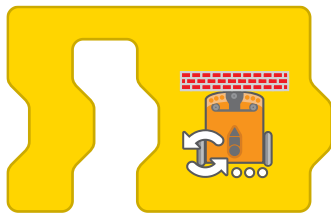
The Edison robot plays a beep, then waits for 0.5 seconds. These blocks are inside a 'loop until round button pushed' block so the beeping will continue until the round button is pushed.



Loop until obstacle

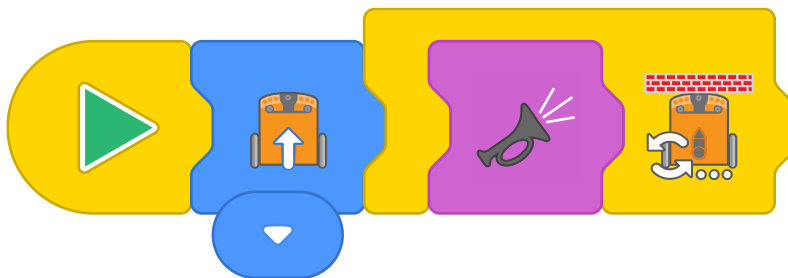
The 'loop until obstacle' block repeats the blocks placed inside of it until, an obstacle is detected.

This is the 'loop until obstacle' block:



Example program

The Edison robot is set to drive forward, then the program enters the 'loop until obstacle' block. The beep block inside the 'loop until obstacle' block causes Edison to repeatedly play the beep sound until, an obstacle is detected.



End program

The 'end program' block is an optional block that can be used to ensure that the program finishes.

This is the 'end program' block:



Example program

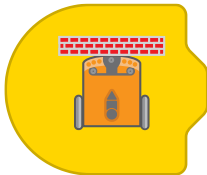
The Edison robot beeps repeatedly forever. However, when the round button is pushed, the program jumps to the 'start when round button pushed' block. The program then ends and the Edison robot stops beeping.



Start when obstacle detected

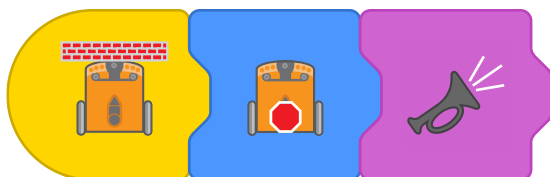
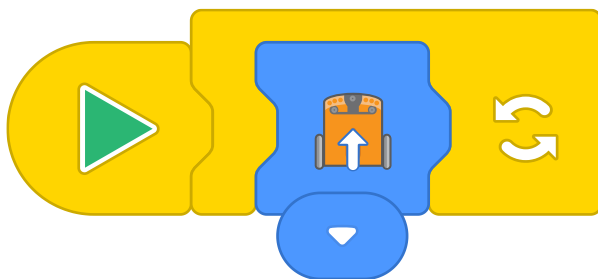
The 'start when obstacle detected' block causes the program to jump to this block when an obstacle is detected.

This is the 'start when obstacle detected' block:



Example program

The Edison robot is set to drive forward inside a 'loop forever' block. When an obstacle is detected, the program jumps to the 'start when obstacle detected' block. The 'stop driving' block stops the Edison from driving and then plays a beep. While the obstacle is present the robot will remain stationary and will continue to beep. Once the obstacle is removed the Edison robot will drive again.



Start when clap detected

The 'start when clap detected' blocks cause the program to jump to this block when the selected number of claps are detected.

There are two 'start when clap detected' blocks:

Start when one
clap is detected

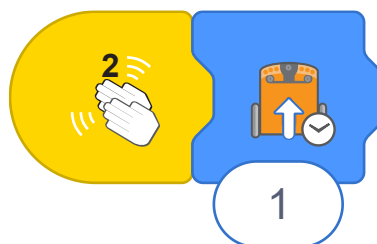
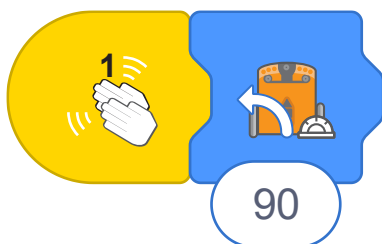


Start when two
claps are detected



Example program

The Edison robot does nothing as the main program is an empty 'loop forever' block. When a single clap is detected, the program leaves the 'loop forever' block and jumps to the 'start when one clap detected' block and Edison turns left 90 degrees. When two claps are detected, the program leaves the 'loop forever' block and jumps to the 'start when two claps detected' block and drives forward for one second.



Start when button pushed

The 'start when button pushed' blocks cause the program to jump to this block when the selected button is pushed.

There are two 'start when button pushed' blocks:

Start when triangle
button is pushed

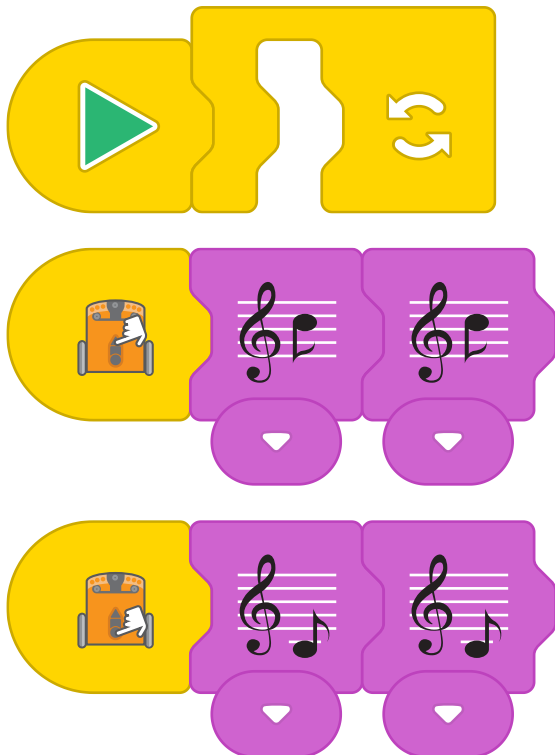


Start when round
button is pushed



Example program

The Edison robot does nothing as the main program is an empty 'loop forever' block. When the triangle button is pushed, the program leaves the 'loop forever' block and jumps to the 'start when triangle button pushed' block and plays two eighth C notes. When the round button is pushed, the program leaves the 'loop forever' block and jumps to the 'start when round button pushed' block and plays two eighth low B notes.



Start when on surface

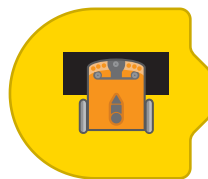
The 'start when on black or white surface' blocks cause the program to jump to this block when the surface changes to either black or white.

There are two 'start when on surface' blocks:

Start when on
white surface

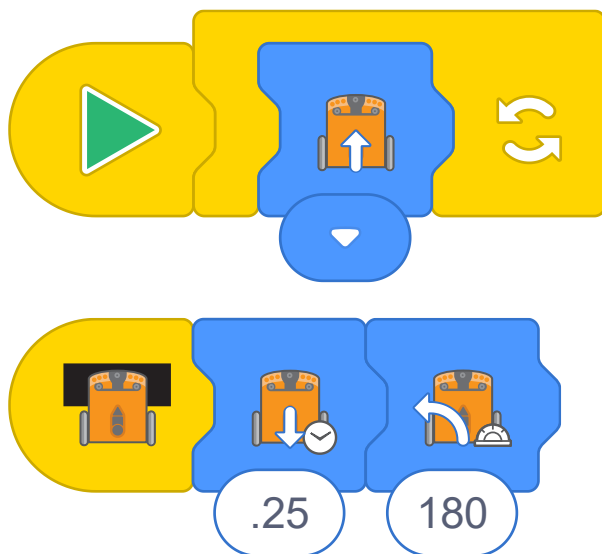


Start when on
black surface



Example program

The Edison robot is set to drive forward inside the 'loop forever' block. When the Edison drives onto a black surface (black line), the program jumps to the 'start when on black surface' block. The Edison then drives backwards for 0.25 seconds then turns 180 degrees around. The program then returns to the 'loop forever' block and the Edison robot is set to drive forward.



Start when message received

The 'start when message received' block causes the program to jump to this block when the selected message colour is received.

There are six 'start when message received' blocks:

Start when blue message received



Start when pink message received



Start when green message received



Start when violet message received



Start when orange message received



Start when indigo message received



Example program

The Edison robot does nothing as the main program is an empty 'loop forever' block. When a blue message is received the program jumps to the 'start when blue message received' block and plays a beep. The program then returns to the empty 'loop forever' block.



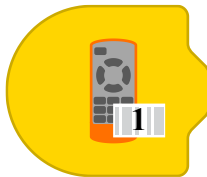
Start when remote command received

The 'start when remote command received' block causes the program to jump to this block when a remote command is received.

For the Edison robot to respond to remote control commands each command from a remote control must be programmed. To do this see the details on the next page.

There are six 'start when remote command received' blocks:

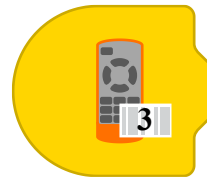
Start when remote command 1 received



Start when remote command 2 received



Start when remote command 3 received



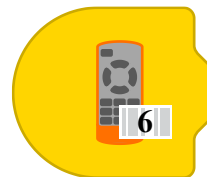
Start when remote command 4 received



Start when remote command 5 received

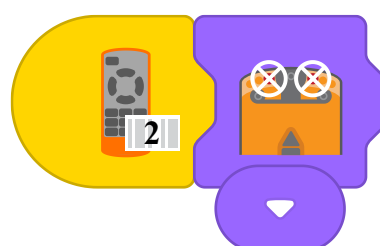
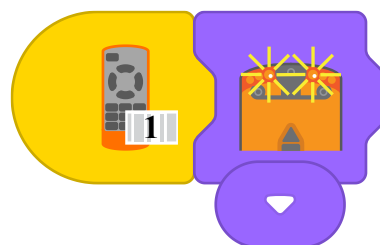
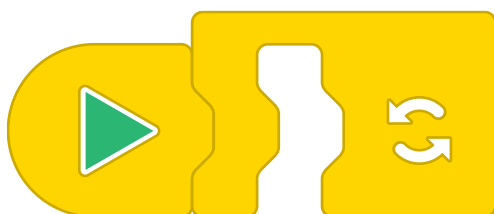


Start when remote command 6 received



Example program

The Edison robot does nothing as the main program is an empty 'loop forever' block. When the remote command 1 is received, both lights turn on. When the remote command 2 is received, both lights turn off.



Setting remote commands

Barcodes are used to teach the Edison V3 each of the six remote control commands.

Steps:

1. Place Edison V3 facing the barcode on the right side of the barcode
2. Press the record (round) button three times
3. Edison drives forward and scans the barcode
4. Press a button on the remote control to set that barcode action

